Design and Implementation of Digital Loop Filter for ADPLL IP Core in 180nm SCL Technology

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Abstract. This project focuses on the design and development of a digital loop filter for an All-Digital Phase- Locked Loop (ADPLL) implemented as an Intel- lectual Property (IP) core in SCL 180nm CMOS technology. The loop filter is a criti- cal component of the ADPLL,it is responsible for stabilizing the system by managing the trade-off between dynamic response and noise suppression. Using Cadence digi- tal design tools, the project entails the end-to-end implementation of the loop filter, including design, synthesis, and post simulation. Key design challenges include opti- mizing the proportional and integral gains to ensure robust performance under vary- ing operating conditions to meet the constraints of the 180nm technology node. This project aims to achieve a high-performance ADPLL design with applications in com- munication systems. The work will be validated through comprehensive simulations, providing insights into the practical implementation of digital loop filters in ADPLL architectures.

Keywords: Digital loop filter, PI control, ADPLL, 180nm SCL, noise suppression, Cadence

1 INTRODUCTION

An All-Digital Phase-Locked Loop (ADPLL) is a crucial component in clock management sys- tems that require phase- aligned clock signals with minimal and consistent jitter. The ADPLL typically comprises three main blocks: a Digitally Controlled Oscillator (DCO), a Time-to-Digital Converter (TDC), and a Digital Loop Filter (DLF). Additionally, a frequency divider may be included in the feedback loop to scale the reference frequency and achieve the desired output frequency. These components serve functions similar to those in an analog PLL, with the place- ment of the DLF highlighted. Since all blocks can be implemented in the digital domain us-
ing synthesizable modules in hardware description languages like Verilog HDL, the system is referred to as an All-Digital PLL.

The TDC in the ADPLL calculates the timing difference between the reference clock (Fref) and the divided output clock of the DCO. A digital signal that frequently

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includes high-frequency noise components is created from this time difference. The DLF is used to filter out these high- frequency components in order to guarantee correct tuning within the DCO's constrained range. Numerous ADPLL architectures have been put forth with the goal of maximizing accuracy and area efficiency

In digital, analog, and radio-frequency integrated systems, phase-locked loops (PLLs) are crucial for producing clock signals with exact frequencies and phases. Because they can take advantage of the digital signal processing capabilities offered by cutting-edge semiconductor technologies, digital PLLs (DPLLs) have become more and more popular in recent years. A time-to-digital converter (TDC) and a digitally controlled oscillator (DCO) are used by DPLLs, in contrast to traditional analog PLLs, to enable the implementation of an entirely digital loop filter. Compared to analog PLLs, this shift offers a number of benefits, such as resistance to circuit flaws, compatibility with deep-submicron CMOS processes, and easier testing and calibration.

In order to reduce noise in the TDC output, a first-order PI loop filter is designed and imple- mented in this paper. Our standalone filter only has a proportional and integral path, in contrast to second-order or higher-order systems. Although TDC and DCO latency may cause higher- order dynamics in full ADPLL systems, our work only considers the loop filter block.

2 PROPOSED ARCHITECTURE OF DIGITAL LOOP FILTER

A proportional Integrator (PI) filter can be used effectively for ADPLL, which is designed in Verilog HDL. Z-domain model of PI filter is shown in Fig (1).

3 Domain PI filter representation

As seen in Fig. 1, a coefficient-based PI filter is designed in this work. Even so, this filter is intended to filter TDC output.

Two main paths make up the suggested digital loop filter: an integrator path with gain KI and a proportional path with gain KL. The 16-bit control signal (Vtune[n]) for the DCO is created by filtering the 16-bit phase error signal (pe[n]) from the TDC fig2). While the proportional term allows for a quick corrective response, the integrator accumulates previous phase errors to correct for long-term drift.

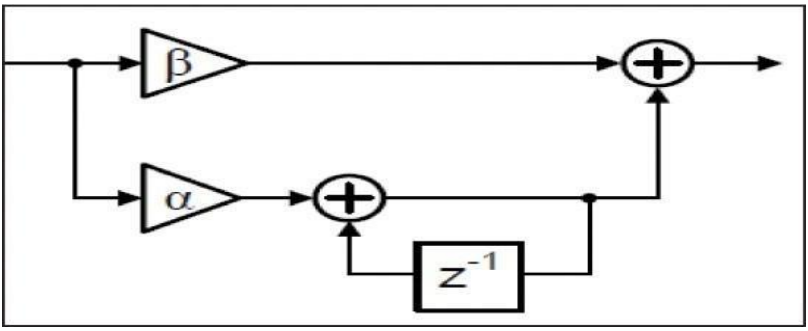


Fig. 1. loop filter Z-domain Architecture

Transfer Function in the Z-Domain:

KL + (KI * z) / (z - 1) = H(z) (1)

This transfer function's first-order behavior is confirmed by the presence of a single zero and a single pole at z = 1.

The difference Equations:

int[n] = int[n-1] + KI * pe[n] (2) Vtune[n] = KL * pe[n] (3)

The coefficients of the control blocks can be modified through a serial link off the chip. In this design the input data is 16 bits, and filter output the filtered 16 bits data for DCO.

4 IMPLEMENTATION OF PROPOSED DESIGN

A proportional gain block and an integrator block are the two main parts of the suggested digital loop filter, which is implemented using Verilog HDL. The filter drives the Digitally Con- trolled Oscillator (DCO) in the ADPLL system by processing a 16-bit input phase error signal (pe[n]) and producing a 16-bit control signal (Vtune[n]).

When the error signal changes, the proportional path instantly corrects it by multiplying the input phase error by a constant gain coefficient KL. The filter can handle long-term frequency drift because the integrator uses the gain KI to accumulate the error signal over time.This behav- ior is described by the following difference equations

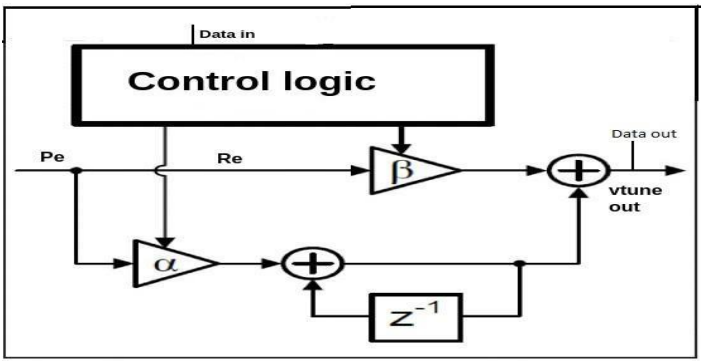


Fig. 2. Interfacing of DLF & Control logic

control equation:

Control Signal(vtune)=Kp · Error+Ki ·.(Error) (4)

These equations are implemented using register-based accumulation, combinational multi- pliers for gain scaling, and summation logic. The filter output is continuously updated with every clock cycle to ensure real-time responsiveness.

This implementation employs fixed coefficient values for KL and KI, which are determined during synthesis. This simplification eliminates the need for runtime configuration interfaces and reduces hardware complexity and area.

This first-order implementation achieves a balanced trade-off between

simplicity and noise suppression capability, making it suitable for applications where deterministic, low-jitter behav- ior is prioritized over runtime adaptability.

The digital loop filter's behavior is modeled using the PI

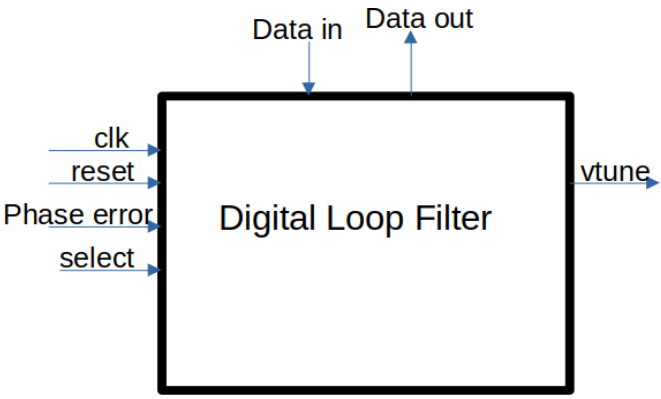


Fig. 3. Block Diagram of Digital loop Filter

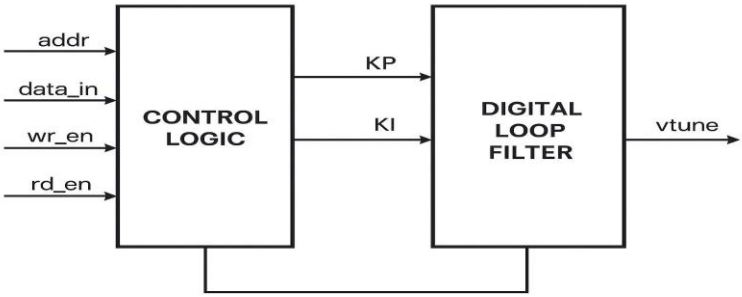


Fig. 4. The control logic Block wrapped to Digital loop filter which can be used for interfacing SPISimulation Results

The Simulation is done using the Cadence EDA tool Ca- dence Xcelium shown in fig(6).Simulation results were ob- tained for the digital loop filter (DLF) using electronic design automation (EDA) tools with a 16-bit input phase error and a 16-bit output control voltage (Vtune). The simulations demon- strated that the DLF effectively suppresses high-frequency noise components in the input phase error signal, ensuring stable and precise control of the digitally controlled oscillator (DCO). The Vtuneoutput showed smooth transitions with minimal ripple, reflecting the filter's ability to maintain phase alignment and mitigate jitter under varying conditions. These results confirm the DLF's capability to deliver high precision and stability within a 16-bit resolution framework, making it well-suited for high-performance digital PLL applications.

By converting the RTL code to Gate level Netlist And Physical Verification is also done.

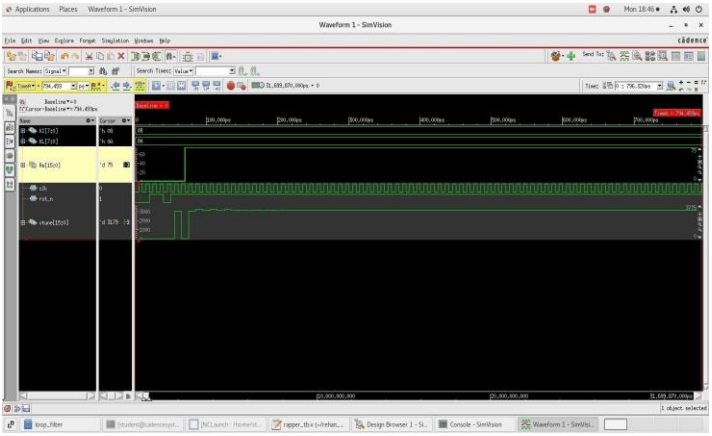


Fig. 5.simulation Results of DLF

5 CONCLUSION

The design and implementation of the digital loop filter for the All-Digital Phase-Locked Loop (ADPLL) in SCL180nm technology have been successfully accomplished, meeting the project's objectives.

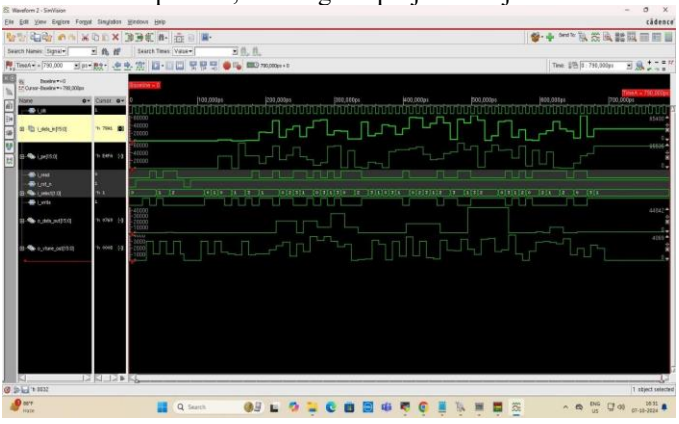


Fig. 6. simulation Results of DLF & control logic

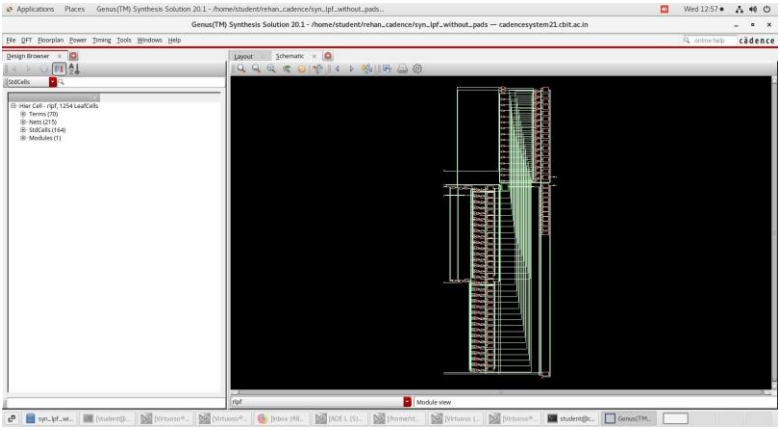


Fig. 7. Synthesized schematic

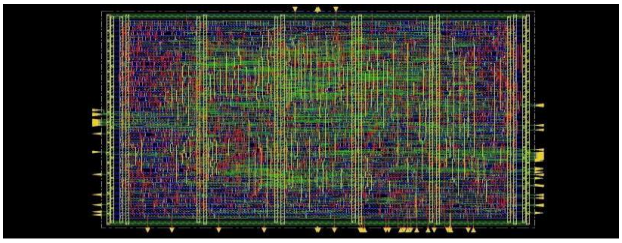


Fig. 8. FInal physical design of DLF

The digital loop filter for the ADPLL in SCL180nm tech- nology was designed and imple- mented using a structured approach, adhering to industry-standard ASIC design flows and lever- aging advanced EDA tools for validation and opti- mization.

The digital loop filter, a crucial component in the ADPLL, demonstrated its effectiveness in suppressing phase noise, mini- mizing jitter, and achieving fast lock times. This ensures reliable performance under dynamic operating conditions. The rigorous design flow, involving RTL implementation, synthesis, and physical design, coupled with extensive validation, ensured that the design adhered to the constraints of the SCL180nm process while achieving optimal power and area efficiency. The successful realization of this project highlights the importance of systematic design method- ologies and the use of advanced EDA tools for developing robust and efficient digital systems.

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